Electrolysis

Move charged particles

1 Electrolyte & electrodes

Ions move only when molten/dissolved. Positive ions go to negative cathode; negative ions to positive anode.

Cathode: gain electrons, reduction. Anode: lose electrons, oxidation. Electrons move in wires, ions in liquid.

2 Molten binary compounds

Metal at cathode; non-metal at anode. No competing water ions.

Molten $PbBr_2$: $Pb^{2+} + 2e^- \rightarrow Pb$; $2Br^- \rightarrow Br_2 + 2e^-$.

Balance atoms and total charge; halogens are diatomic.

Predict the products

3 Aqueous product rules

Water adds H^+ and OH^-. At cathode: H_2 if metal is above hydrogen, otherwise metal.

At anode with inert electrodes: halogen if halide present, otherwise O_2.

NaCl(aq): H_2/Cl_2; $CuCl_2$(aq): Cu/Cl_2; $CuSO_4$(aq): Cu/O_2.

4 Useful half-equations

$2H^+ + 2e^- \rightarrow H_2$. $Cu^{2+} + 2e^- \rightarrow Cu$.

$2Cl^- \rightarrow Cl_2 + 2e^-$. $4OH^- \rightarrow O_2 + 2H_2O + 4e^-$.

Oxidation puts electrons on product side; reduction uses them on reactant side.

Apply & investigate

5 Aluminium extraction

Molten Al_2O_3/cryolite mixture needs lower temperature than pure Al_2O_3. $Al^{3+} + 3e^- \rightarrow Al$. $2O^{2-} \rightarrow O_2 + 4e^-$. Oxygen reacts with carbon anodes → CO_2; replace anodes.

Energy needed for heat and current; carbon cannot reduce aluminium oxide.

6 Aqueous practical

Hypothesis: changing dissolved compound changes products. Use inert electrodes/DC; compare solutions at same voltage, time, concentration and electrode setup.

Test H_2: pop; O_2: relights glowing splint; Cl_2: bleaches damp litmus. Record metal deposits too.

Repeat; goggles, ventilation for chlorine; switch off before adjusting electrodes.